Quiz 1

CS 102 Computer Programming I · Fall Semester 2026
Week 4 · Duration: **35 MINUTES** · Version **D**

New Uzbekistan University

FULL NAME: _____ ID: _____ GROUP: _____

Rules. No notes, phones or computers. Write answers *only* in the space given. For fill-in-the-gap questions write exactly one piece of C++ code in each numbered gap — a keyword, an operator, a type or an expression. Assume every code fragment is inside `main` with `#include <iostream>` and `using namespace std;`.

Section	Points	Score
A Multiple choice	30	
B Fill in the gaps	40	
C Trace and write	30	
Total	**100**	

Section A: Multiple Choice (30 points)

Circle **one** answer for each question.

A1. (6 points) The hexadecimal literal `0x41` has the decimal value: A. 41 B. 81 C. 56 D. 65

A2. (6 points) What does `cout << 8 / 3 * 1.5 + 8 % 3;` print? A. 6 B. 5 C. 4 D. 6.5

A3. (6 points) What does this code print?

```
int x = 4;
if (x < 0 || ++x > 4) {
    cout << "ok ";
}
cout << x;
```

A. `ok 4` B. `ok 5` C. `4` D. `5`

A4. (6 points) Which expression does **not** evaluate to `1.5`?

A. `3 / 2.0`

B. `static_cast<double>(3) / 2`

C. `3.0 / 2`

D. `static_cast<double>(3 / 2)`

A5. (6 points) What does this print? `char c = 'a' - 1 + 5; cout << c << (c - 'a');`
Hint: in the ASCII table, `'a'` *= 97.* A. `e4` B. `e101` C. `1014` D. `d4`

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Section B: Fill in the Gaps (40 points)

Each numbered gap is worth 2 points. Every program must compile and produce the output shown. Write your answers in the boxes under each program.

B1. (10 points) **Hours to weeks, days, hours.** Input `200` must print `1 w 1 d 8 h`.

```
#include <(1)_____>
using namespace std;
int main() {
    int total;
    cin (2)_____ total;
    int w = total (3)_____ 168;              // 168 hours in a week
    int d = (total (4)_____ 168) / 24;
    int h = total (5)_____ 24;
    cout << w << " w " << d << " d " << h << " h" << endl;
    return 0;
}
```

(1) _____ (2) _____ (3) _____ (4) _____ (5) _____

B2. (10 points) **Average temperature.** Level `H` for an average of 25 or more, `W` for 10 or more, otherwise `C`; averages below 0 or above 60 are invalid. Input `20 25 31` must print `Average: 25.3333, rounded: 25, level: H`.

```
int t1, t2, t3;
cin >> t1 >> t2 >> t3;
double avg = (6)_____(t1 + t2 + t3) / 3;
if ((7)_____) {                // outside 0..60
    cout << "Invalid" << endl;
} else {
    (8)_____ level;
    if (avg (9)_____ 25) {
        level = 'H';
    } else if (avg >= 10) {
        level = 'W';
    } else {
        level = 'C';
    }
    cout << "Average: " << avg << ", rounded: " << (10)_____(avg + 0.5)
         << ", level: " << level << endl;
}
```

(6) _____ (7) _____ (8) _____ (9) _____ (10) _____

B3. (10 points) **Largest of three.** Input `3 9 5` prints `Max: 9`; `4 4 4` prints `All equal`.

```
int a, b, c;
cin >> a >> b >> c;
int mx = a;
if (b (11)_____ mx) {
    mx = b;
}
if (c > (12)_____) {
    mx = c;
}
bool allEqual = (a == b) (13)_____ (b == c);
if ((14)_____) {
    cout << "All equal" << endl;
} (15)_____ {
    cout << "Max: " << mx << endl;
}
```

(11) _____ (12) _____ (13) _____ (14) _____ (15) _____

B4. (10 points) **Savings plan (watch the total!).** A monthly deposit is up to 10^6 so'm for up to 10^4 months. A plan is *rich* if the total is at least 10^9 so'm and *long* if it lasts at least 120 months. Input `900000 3000` must print `Both`.

```
int deposit, months;
cin >> deposit >> months;
(16) _____ total = static_cast<long long>(deposit) * months;
bool rich = total >= 1000000000;
bool longTerm = months >= 120;
if (rich (17) _____ longTerm) {
    cout << "Both" << endl;
} else if ((18) _____) {
    cout << "Rich" << endl;
} else if (longTerm) {
    cout << "Long" << endl;
} (19) _____ {
    cout << "Neither" << endl;
}
```

(16) [ ] (17) [ ] (18) [ ] (19) [ ]

(20) Why would `int total = deposit * months;` print `Long` for the input above?

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Section C: Trace and Write (30 points)

C1. (15 points) Write **exactly** what this fragment prints (mind the spaces).
Hint: in the ASCII table, 'a' = 97 (and 'A' = 65).

```
int a = 10, b = 4, c = 0;
double d = a / b + a % b / 4.0;
if (a > b)
    if (c != 0 && a / c < 5)
        cout << "M";
else
    cout << "N";
cout << " " << d;
char ch = 'z' - b;
if (ch >= 'a' && ch <= 'z')
    ch = ch - 32;
cout << " " << ch << " " << (ch < 'W') << endl;
```

C2. (15 points) **Cooking time.** A dish is started at `hh mm`; preparing takes `p` minutes and baking `k` minutes. Write a program that reads `hh mm p k` and prints when the dish is ready as hours and minutes. Do **not** wrap around at 24: hours simply keep growing. Use only `int`, `/` and `%`.

```
18 40 35 90   → 20 h 45 min
23 30 60 60   → 25 h 30 min
9 0 15 45     → 10 h 0 min
```

END OF QUIZ